Final Year Project Report

Interim Report

Browser-Based Game for Teaching Cyber Security Concepts

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A report submitted in part fulfilment of the degree of

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Declaration

This report has been prepared on the basis of my own work. Where other published and unpublished source materials have been used, these have been acknowledged.

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Table of Contents

Final Year Project Report 1

Browser-Based Game for Teaching Cyber Security Concepts 1

Elizabeth Eardley 1

Table of Contents 2

Chapter 1: Introduction 3

1.1 Game description 3

1.2 Aims and Objectives 3

1.3 Educational Objectives 3

1.4 Quest descriptions 3

1.4.1 What is cyber security 3

1.4.2 Understanding cyber risk 3

1.4.3 Common cyber attacks 3

1.4.4 CIA 3

1.4.5 IAAA 3

1.4.6 People and security 3

1.4.7 Defensive mechanisms 3

1.4.8 Operating systems 3

1.4.9 Security controls 3

1.4.10 Cryptography basics 3

1.5 Literature survey 3

Chapter 2: Planning and time scale 3

2.1 Iterative development approach 3

2.2 Use of version control 3

2.3 Modular quest design 3

2.4 Future testing strategy 3

2.5 Future Evaluation through user playtesting 3

Chapter 3: Summary of completed work 3

3.1 Game world & maps 3

3.2 Narrative design 3

3.3 Game mechanics 3

3.4 Educational mappings 3

3.5 Completed work 3

3.6 Project diary 3

Bibliography 3

Chapter 1: **Introduction**

1.1 Game description

In a distant future, the world of Cipheria stands on the edge of destruction. Once a harmonious realm powered by the Motherboard, a living network that sustains life and preserves the memories of every citizen, it now descends into chaos as powerful forces exploit the cracks in its defences, corrupting the Motherboard and twisting the very world it was created to protect.

The player takes on the role of a young apprentice guardian, tasked with defending Cipheria and defeating its enemies alongside one of Cipheria's legendary elders.

Through this journey, the player will learn the art of cybersecurity through story-driven quests, immersive puzzles, and strategic mini-games. Each quest transforms a real-world cybersecurity concept into an interactive challenge—teaching players how to protect systems, identify risks, and outsmart attackers while restoring balance to Cipheria.

1.2 Aims and Objectives

The Aims of this project are the development of a browser-based, story-driven game to teach cybersecurity concepts through quest-like challenges. Each quest will reflect a cyber security concept in a metaphorical way in hopes of transforming cyber security education into an engaging and meaningful learning experience accessible to anyone, especially those who are new to the field.

1.3 Educational Objectives

The goal of the game is to teach foundational cybersecurity principles through interactive, story-driven gameplay. Each quest transforms a real-world cybersecurity concept into a challenge using metaphors, allowing players to learn by doing rather than learning passively.

By completing all the quests, players will understand the following cybersecurity concepts:

1. What is cybersecurity - understanding the importance of protecting digital assets
2. Understanding cyber risk - identifying vulnerabilities, threats, exploits and the impact each risk has
3. Common cyber attacks - recognising each attack and its countermeasures
4. CIA triad
5. IAAA principles
6. People and security - recognising social engineering and practising strong password hygiene
7. Defensive mechanisms - understanding firewalls, antivirus systems, encryption and intrusion detection
8. Operating systems - learning how different systems have their own vulnerabilities
9. Security controls - distinguishing between technical, administrative and physical controls
10. Cryptography basics

Each concept will be introduced progressively, reinforced through challenges and consolidated through boss encounters, which will test the player's comprehension.

The game will follow a structured educational loop designed to introduce, reinforce and assess the player's learning throughout the gameplay.

- Introduction phase - concept introduced via story dialogue and a simple quest. The player will learn through guided tutorials or easy challenges.
- Consolidation phase - concepts are reinforced through progressively harder puzzles.
- Final boss phase - Each concept will have a boss challenge, which will test the mastery of that concept. The player's understanding is tested.

1.4 Quest descriptions

1.4.1 What is cyber security

The player is welcomed to Cipheria, a futuristic world powered by a massive Tower where the Motherboard resides. The Motherboard is the heart of Cipheria - a powerful source made up of three crystals that keep the world alive and connected. But dark forces seek to steal the crystals, for whoever controls the Motherboard controls the entire world.

In this level, the player must climb the Tower to reach the Motherboard at its peak, defending it from waves of enemy robots and drones trying to break in and steal its power.

This quest will reflect on the concept that cyber security essentially is protecting valuable assets from threats

1.4.2 Understanding cyber risk

After reaching the top of the Tower, the player enters the Motherboard, the heart of Cipheria. Inside, the player must explore its inner systems to locate and repair areas of damage before they worsen.

- **Vulnerabilities:** Cracks or damage within the Motherboard's structure.
- **Exploits:** Enemies squeezing through those cracks to cause more harm.
- **Threats:** The enemies themselves attacking the system.
- **Threat Actor:** A boss enemy commanding and coordinating the attacks.

As the player navigates the level, a risk percentage appears beside each damaged area, showing the chance of an enemy breaking through if it isn't repaired in time.

At the end of the quest, the player exits the Tower and speaks with Cipheria's citizens, witnessing firsthand how the attacks have affected their world - linking the player's actions to real-world consequences.

This quest will give the learning an understanding of the meaning of each of the following terms: vulnerabilities, exploits, threats, threat actor, risk and impact.

1.4.3 Common cyber attacks

After protecting the Motherboard, the Elder sends the player to the Null Domain - a dangerous region where chaos and corruption reign. The Elder explains that to become a true Guardian, the apprentice must learn to recognize and defend against the most common cyber threats that endanger Cipheria.

Inside the Null Domain, the player faces a series of encounters that represent real-world cyberattacks:

- **Phishing:** A trickster NPC disguises themselves as the Elder and tries to deceive the player into giving away the keys to the Tower to gain access to the Motherboard.

- **Malware:** Swarms of slime-like creatures that multiply each time they are touched, spreading digital corruption through the area.

- **DDoS (Distributed Denial of Service):** Endless waves of small drones that overwhelm the player's defenses through sheer numbers.

- **Ransomware:** A cyborg enemy steals and locks away the player's sword, forcing them to either solve a puzzle or surrender the keys to the Motherboard.

The player must face each type of attack three times, with each encounter changing its format or appearance to test the player's ability to recognize threats in different forms. In the final stage, the player will need to match each cyberattack name to the threat in front of them, reinforcing pattern recognition and situational awareness.

After this quest the learning should be able to identify common cyberattacks and their countermeasures, emphasizing that these threats can appear in different forms and must be recognized through their behavior and intent, not just their appearance.

1.4.4 CIA

After returning from the Null Domain, the player arrives back at the main tower to find that the 3 crystals that power the Motherboard have been stolen and tampered with. The Elder explains that these crystals represent the CIA - Confidentiality, Integrity and Availability - and that they are essential to keep the balance in Cipheria. The player must locate and restore each crystal:

- **Confidentiality crystal** - hidden deep within a dark alley of Cipheria's lower districts. To find, the player must earn the trust of local citizens who may have witnessed anything. The player will engage in dialogue puzzles, choosing responses to build trust and demonstrate discretion. Fake informants attempt to trick the player into revealing too much information or sharing secrets, enforcing that not everyone can be trusted. The player must use a passkey to prove their identity before the citizens reveal their clues.
- **Integrity crystal** - still located within the Motherboard, but has been tampered with. The player must analyze it and restore it to its original state. If the player fails to fully repair it, the virus begins to spread, infecting the Motherboard and as a consequence Cipheria
- **Availability crystal** - Trapped inside a chamber blocked by endless waves of drones. The player must restore access by clearing the enemies and placing it back on its pedestal to make it available again.

The principles of the CIA triad are central to cyber security, after this quest the learning will understand that protecting all 3 ensures the stability and security of a system.

1.4.5 IAAA

After heroically restoring balance to Cipheria, the Rulers of Cipheria invite the player to their fortress to honor their achievements. Enclosed by three walls, the fortress represents layers of security.
Before entering, the player receives a Cyber Seal along with an official invitation, both of which will be used throughout the entry process. To gain access, the player must successfully pass through each security layer, each representing a stage of the IAAA framework:

- **Identification (First Wall):** Upon arriving at the outer gate, the guards stop the player and ask for their name. They check it against the official visitor list to confirm the player's identity.
- **Authentication (Second Wall):** Once past the first wall, the player must present their Cyber Seal - a unique, encrypted symbol given in the invitation. Guards verify that the seal matches the record on the visitor list. Only if it matches will the player be allowed to proceed.
- **Authorization (Third Wall):** At the innermost gate, the player is asked to show proof of their invitation, confirming they are not only identified and authenticated but also authorized to enter the fortress and meet the rulers.

After successfully passing all three walls, the player is assigned a Scribe, that records all actions during their stay inside the fortress — representing the final principle: **Accountability**.

The quest will reflect the IAAA principles, their importance and how each layer works together to secure access, verify identity and ensure each action is accountable within a secure system.

1.4.6 People and security

After having met the rulers of Cipheria, the player makes their way back to the main tower. On their journey they will interact with citizens, who seek guidance and protection from the newly arising threats. However, not everyone is who they say they are. Some citizens are tricksters in disguise, using deception and persuasion to try convince the player to reveal some of the citizens' passkeys or the player's cyber seal. Other citizens will genuinely ask for help, explaining that they need a guardian to create strong passkeys for their homes. The player must craft secure passwords for them.

Sometimes people are often the weakest link in cybersecurity and that being aware of social engineering and reckognising it is crucial.

1.4.7 **Defensive mechanisms**

As the player returns to the main tower, the Elder tasks them with building and strengthening the tower's defences to protect the Motherboard from future attacks. The player must craft the following defences:

- Firewall - a barrier surrounding the tower blocking enemy drones or robots allowing only trusted entities to enter.
- Antivirus - a maintenance robot which roams the tower's interior cleansing any viruses or corruption.
- Encryption - The door to the Motherboard will only show itself if a special sequence of characters is entered.
- IDS - a radio like creature, which monitors for suspect behaviour and announces if enemies manage to sneak in.

This quest will go through the importance of defensive mechanisms and that placing multiple layers of protection is good practice for maintaining long-term protection.

1.4.8 Operating systems

Cipheria is made up of several kingdoms, each ruled by a unique operating system and its own digital language:

- **The Purple Kingdom – Linux**
- **The Green Kingdom – Windows**
- **The Yellow Kingdom – Mac**

An enemy hacker sends the player a series of encrypted puzzles, each written in one of these kingdom's languages. To decode them, the player must travel to the corresponding kingdom, where the language becomes readable.

In each kingdom, the player must translate and solve puzzles that reveal fragments of the hacker's plan. Once all puzzles are solved, the player can trace the source of the attack and catch one of Cipheria's evil masterminds.

The learning will gain an understanding that each operating system have unique structures, languages and vulnerabilities and how it is important to understand how each system differs.

1.4.9 Security controls

After successfully completing every challenge and proving their heroic nature, the Elder assigns the player one final test before officially crowning them as a Guardian of Cipheria. The player must design and build a new Abbey for the Elders, implementing three key categories of security controls to protect it from any attacks.

- Technical controls: construct walls, moats, and defense towers to protect the Abbey's systems and prevent unauthorized access.
- Administrative controls: create a set of rules and policies for those who live and work within the Abbey.
- Physical controls: hire guards and install locks on gates and doors, representing physical barriers that protect against any intrusion.

Once the construction is complete, the Abbey's security will be tested in a mock attack simulation. If the player's security design is strong, the Abbey will pass the test. If too few controls are in place, the player must reattempt until the Abbey passes the test.

The learner will understand how each category of security controls work together to create a resilient defence system.

1.4.10 Cryptography basics

A mysterious message appears on the Motherboard - a distress signal written in a way which is unreadable. The Elder believes it could come from one of the missing guardians who travelled into the Null Domain to try defeat some of Cipheria's worst criminals. The player must decode the message and reveal its meaning. Once decrypted, the message confirms the guardian is still alive and in danger. To communicate, the player must encrypt new messages using the same cipher. This starts a coded conversation between the player and missing guardian, with each exchange revealing more about:

- What happened during the guardian's failed mission in the Null Domain
- The identities of the final bosses
- The location of the missing guardia

Each message becomes more complex to decode, and different cyphers may be used to ensure that the conversation is secure and confidential.

This quest will go through the foundations of cryptography, including encryption, decryption and secure communication.

1.5 Literature survey

Game-based learning combines entertainment and education to create immersive and interactive experiences that promote active learning. Recent studies highlight that educational games have become more recognised as effective learning tools across many disciplines.

In a study conducted by Begg, Dewhurst and Macleod (2005), two narrative-based scenarios from the University of Edinburgh's medical curriculum demonstrated how problem-based learning principles can transform traditional instruction. In these activities, students respond to evolving situations presented in an engaging, game-like style uncommon in standard university coursework. The study found that successful game-informed learning followed several key principles. First, the game needed a compelling backstory and context to provide emotional entry points and relevance

for the learner. Continuous feedback ensured that learning felt rewarding and progress was visible. Learners were also given opportunities for safe failure, allowing them to make mistakes and try again without real-world consequences. Players adopted meaningful roles that shaped their behaviour and engagement within the learning environment, while emotional connections made the experience more memorable and authentic. Finally, consequential decision-making—where players' actions directly influenced outcomes—helped create a strong sense of motivation and active participation in the learning process.

Similarly, James Paul Gee (2004) argued that games could be powerful learning systems, which enable learners to "learn by doing" through active co-creation of their experiences. Video games, in general, combine motivation, challenge and feedback in ways that would make learning more fun and engaging. Additionally, games also encourage exploration, experimentation and productive failure establishing a very healthy relationship to continuous learning and improvement.

From a broader perspective, de Freitas (2018) observed that research on the use of games as educational tools is fragmented across many disciplines such as education science, game studies, neuroscience and information science - resulting in inconsistent terminology and methodology. She also identified the ongoing challenge of aligning fun and playability with learning outcomes, putting an emphasis on the need for thoughtful design and planning that balances fun and education. De Freitas also argued that many studies highlight that game-based learning show positive cognitive and behavioural impacts to the learner, improving knowledge retention, motivation and skill development.

Building further on these theories of learning, the Dreyfus Model of Skill Acquisition (1980) offers insight into the gradual development of expertise over time. The model proposes five stages of skill acquisition where each stage represents a deeper understanding on the subject being learnt. The five stages which the model proposes are: novice, advanced beginned, competend, proficient and expert. Applying this model within the game would ensure scaffolding learning, where players can build confidence, competence and evolve their abilities over time.

Chapter 2: **Planning and time scale**

2.1 Development approach

The project adopts an iterative development approach, allowing functionality to be developed, reviewed, and refined incrementally. This approach was chosen due to the evolving nature of the project, where gameplay mechanics, narrative elements, and educational content influence one another. By working in short development cycles, the project can adapt to new requirements or design changes without requiring extensive rework.

Each iteration focuses on a small set of objectives, such as implementing a quest, refining dialogue, or improving interaction mechanics. This allows progress to be evaluated regularly and ensures that the project remains aligned with its educational aims throughout development.

2.2 Use of version control

Git is used as the version control system for the project, supporting good software engineering practice. Version control enables changes to be tracked over time and allows experimentation without risking loss of previous work. Regular commits are made to document development progress and to maintain a clear history of design and implementation decisions.

The use of Git also supports modular development, as individual features or quests can be developed and refined independently before being integrated into the main codebase. This approach improves reliability and maintainability as the project grows in complexity.

2.3 Quest design

The game is designed around a modular quest structure, where each quest represents a cybersecurity concept. Quests are implemented as independent components with clearly defined entry points, dialogue, and completion conditions. This modular design allows new quests or learning topics to be added with minimal impact on existing functionality.

From a planning perspective, modularity supports parallel development of narrative, mechanics, and educational content. It also allows the scope of the project to be managed effectively, as individual quests can be prioritised or deferred depending on time constraints while maintaining overall project coherence.

2.4 Future testing strategy

Although the primary focus of the first term is on design and early implementation, testing has been considered in the planning process. Future testing will focus on both technical correctness and educational effectiveness. Within the Phaser and TypeScript environment, testing will include verifying player input handling, collision detection, dialogue triggers, and quest completion logic.

Manual playtesting will be used initially to evaluate gameplay flow, usability, and clarity of educational metaphors. As development progresses, structured testing scenarios will be introduced

to ensure that new features do not introduce regressions. This testing strategy supports maintainability and reliability as the project continues into later stages of development.

2.5 Future Evaluation through user playtesting

In addition to technical testing, the project plan for next term includes user-based playtesting to evaluate whether the game is effective in teaching cybersecurity concepts. This form of evaluation focuses on how real users interact with the game and whether the game is effective in teaching the concepts it introduces.

Participants will be asked to play selected sections of the game and provide feedback on their experience. This feedback will focus on clarity of objectives, engagement, ease of use, and understanding of the cybersecurity concepts presented. Observing how players respond to in-game metaphors and narrative explanations will help determine whether the educational mappings are effective or require refinement.

Playtesting sessions will also be used to identify usability issues, such as unclear controls, confusing dialogue, or ineffective feedback mechanisms. Insights gained from these sessions will inform iterative improvements to both gameplay mechanics and educational design.

This approach ensures that the final product is not only functional but also meets its primary goal of supporting learning through interactive gameplay.

Chapter 3: **Summary of completed work**

3.1 Game world & maps

The game world was designed and created using Tiled, a map editor used for 2D game development. It was selected because of its ease of use and flexibility. Tiled allows tile-based maps to be created efficiently while also supporting the placement of objects directly within the map. These objects can then be used to represent interactive elements within the game, such as non-player characters, doors, or triggers for enemies or dialogues.

An important advantage of using Tiled was the ability to define custom properties for both maps and individual objects. These properties can be used to store information such as interaction types, identifiers, or behavioural flags (for example whether a door was open or locked), allowing game logic to be driven by data rather than hard-coded values. This approach improves maintainability and makes it easier to modify or extend gameplay features as development progresses.

The maps were designed to link together logically, forming a game world that supports exploration and progression. A modular approach was adopted, enabling new areas to be added in the future without disrupting existing content. Tile sets with vibrant colours were chosen to create a visually engaging environment, helping to make the game feel less intimidating and more enjoyable, particularly for players who may be new to cybersecurity concepts.

Overall, the use of Tiled contributed to a flexible and extensible level design workflow, supporting both the technical requirements of the project and the goal of creating an engaging educational game world.

3.2 Narrative design

Narrative design plays a central role in the project, as it is used to contextualise cybersecurity concepts and provide continuous feedback to the player. The game is set within a fictional world where abstract technical ideas are represented through characters, environments, and events, making them more accessible and easier to understand.

A key narrative element is the presence of an Elder character, who acts as a guide throughout the game. The Elder introduces quests, explains objectives, and provides feedback as the player progresses. Dialogue from the Elder is used not only to advance the story but also to explain cybersecurity concepts using metaphors that align with the game world. For example, digital assets, threats, and vulnerabilities are represented through physical objects, enemies, and environmental damage, helping players form intuitive mental models of otherwise abstract ideas.

The Elder comments on the player's actions, reinforcing correct decisions and providing guidance on what just happened. This approach supports learning by allowing players to reflect on their actions in real time and adjust their understanding without breaking immersion.

By embedding feedback and instruction within the narrative, the game avoids interrupting gameplay with explicit explanations or external tutorials. This design choice helps maintain engagement while ensuring that educational content is clearly communicated. Overall, the narrative design supports both the gameplay experience and the educational objectives of the project by linking story progression directly to learning progression.

3.3 Game mechanics

The gameplay mechanics form the foundation of player interaction with the game world. Core mechanics were implemented with extensibility in mind, allowing additional features or quests to be added incrementally without major changes to the existing code.

Player mechanics include basic states such as idle, running, and attacking. These states are controlled through keyboard input. Animations and state transitions were implemented to maintain responsiveness and visual clarity, ensuring the player receives immediate feedback on their actions.

Enemy behaviour is designed to challenge the player while reinforcing cybersecurity concepts. Basic enemies are programmed to follow the player. More complex enemies appear in specific quests to represent real-world cyber threats, such as ransomware or malware, creating gameplay scenarios that directly link to the educational content.

Dialogue systems are an essential part of the game mechanics. Dialogue is categorised into four types: introductory dialogue, quest completion dialogue, interaction dialogue, and tutorial dialogue. These dialogue types are triggered contextually to guide the player through quests, explain concepts using metaphors, and provide feedback on performance. This structure ensures that narrative and educational objectives are tightly integrated with gameplay.

Overall, the game mechanics were designed to be simple yet flexible, supporting incremental development and future feature additions. By linking player actions, enemy behaviour, and dialogue with educational objectives, the mechanics reinforce learning while maintaining an engaging gameplay experience.

3.4 Educational mappings

Educational mappings were a central consideration in the design of the game, ensuring that gameplay mechanics and narrative elements directly support the intended cybersecurity learning outcomes. Rather than presenting concepts through explicit technical explanations, the project uses metaphors to translate abstract cybersecurity principles into intuitive, in-game experiences.

Metaphors are used to explain the concepts such as the CIA and vulnerabilities. Physical barriers, guards, and layered walls are used to represent authentication, authorisation, and accountability, allowing players to understand these principles through interaction rather than memorisation. Similarly, defensive technologies such as firewalls, antivirus systems, and intrusion detection systems are represented by in-game constructs that block, detect, or remove enemies.

This metaphor-driven approach supports experiential learning by allowing players to learn through observation, experimentation, and feedback. By interacting with these metaphors, players can develop mental models that closely mirror real-world cybersecurity concepts. This design aligns with established theories of game-based learning, which emphasise learning by doing and the importance of contextualised feedback.

Overall, the use of educational mappings ensures that gameplay, narrative, and learning objectives remain tightly integrated. This approach helps maintain player engagement while ensuring that educational content is delivered in a meaningful and accessible way.

3.5 Completed work

Quests 1 and 2 were completed to a good standard. Both quests include appropriate game assets, full dialogue, and are fully linked within the game world, allowing the player to progress smoothly through these sections.

Quest 3 is functional; however, some of the intended quest objectives are currently missing. For example, the puzzle required to retrieve the player's sword in the ransomware room has not yet been fully implemented. This quest therefore requires further development and refinement in the next term.

Quests 5 and 6 are fully linked within the game world and include dialogue; however, the associated game assets have not yet been added. These quests are structurally in place but require visual and gameplay enhancements to reach the intended standard.

A base project structure was created, including managers for quests, maps, and dialogue. This structure was designed to support modular development, making it easier to add new content and functionality in future stages of the project. Establishing this foundation early was a deliberate decision to support long-term development and maintainability.

During the project, there were occasions where AI tools, specifically Claude AI (Sonnet 4.5), were used to assist with debugging sections of code when error messages were not immediately understood. Any code generated during this process was removed afterwards, and all final solutions implemented were my own. Areas where code generated by AI was not removed have been clearly marked in the project files, along with the prompts used.

3.6 Project diary

29-09-2025 to 12-10-2025	- Researched and reviewed serious games for cyber security - Researched learnning strategies - Created project plan
13-10-2025 to 19-10-2025	- Created a document outlining game mechanics, educational objectives and user flow for the prototype - Familiarised with Phaser
20-10-2025	- Cloned git repository to local machine - Set up the repository structure - Project plan added to documents - Game mechanics, educational objectives and userflow document added to documents

21-10-2025	- Ethics questionaire completed - main character sprite sheet generated
30-10-2025	- Created issue board - added issues and their description so far in to-do board and the sprint to-do list - Created milestones on gitlab and assigned issues to each milestone
week 5	- Map creations for quests 1 till 6 - found assets
week 6	- Implementation for Player - started adding maps to project repo - Ground work for project structure (scenes, managers, config files)
week 7	- Started linking maps together - Ground work for project structure (scenes, managers, config files)
week 8	- Quest configuration update
week 9	- Enemy implementation
week 10	- debugging - start setting up the quests
week 11	- coding the quests and dialogues

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